



# United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam, Trimester: Spring 2025

Course Code: CSE-3521, Course Title: Database Management Systems

Total Marks: 40, Duration: 2 hours

**Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.**

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1. a) Consider a schema **R(A,B,C,D,E,F,G)** with the following set of functional dependencies.

**A → C**

**AB → DE**

**CF → D**

**C → GFD**

**D → F**

**G → A**

3

+3

+2

=8

i) Find out all possible **candidate keys**.

ii) Compute the **minimal cover** of the set of functional dependencies.

iii) Based on the **minimal cover**, examine each functional dependency and **determine** whether the relation is in **2NF** or not.

- b) Suppose we have a relation named **R**, having attributes as **P, Q, R, S, T** and **U**. The relation **R** is having the following functional dependencies,

2+

2=

4

**QR → S**

**S → T**

**TU → P**

**P → Q**

i) Considering the relation with functional dependencies, find if **PR → T** is valid or not using **Armstrong's axioms**. [mention the axioms applied in each state]

ii) If we decompose the relation in **R1(P, Q)** and **R2(S, T)**, **R3(P, R, S, U)** then **determine** whether this decomposition is **lossless** or **lossy**.

2. Consider the following hash function - 10

$\text{hash}(\text{Search\_key}) = \text{position of the letter in the alphabet}$

For example: 'A'  $\rightarrow$  1, 'Z'  $\rightarrow$  26

and the given search keys:

**A, P, T, J, X, G, K, R, E, I**

Considering the **LSB mapping of the hash values**, insert the search keys one by one into the extensible hash structure using the given hash function. Assume that the bucket capacity is 2 and the initial local and global depths are 1.

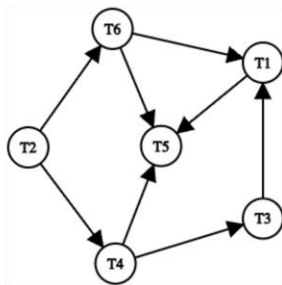
3. a) Clearly explain the difference between indexing and hashing in databases. In what scenarios is each method preferable? 3

b) Consider the following data coming from left to right. Construct a **B+ Tree of order 3** for the incoming data and show all necessary steps while inserting. 7

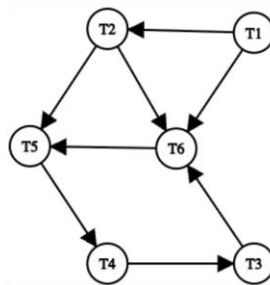
5, 12, 7, 20, 3, 9, 15, 1, 18, 25

4. a) Explain the dirty read (uncommitted dependency) problem with an example in the context of transactions. 2  
 b) Consider the following precedence graphs and determine whether the transactions are serializable. If they are serializable, find all the serial schedules; otherwise, explain why they are not serializable. 6  
8

i.



ii.



iii.

